Module 4: Sampling and Estimation Review

Select, calculate, interpret, qualify

Table of contents

Integrated tasks 1
Bridge to hypothesis testing 1

Integrated tasks

Use the checked cases from Lessons 1–4; no new business observations are introduced. Submit a concise report with a method choice, calculation, interpretation, and limitation for each task.

1. **Selection and bias:** Explain why a random sample from an incomplete EAI employee frame may not represent all employees. Distinguish parameter, estimate, realized error, and SE.
2. **Sampling distribution:** For EAI salary samples of n=100, compute the approximate and corrected SE for N=2500. Contrast the SE with the $4,000 population SD.
3. **Mean interval:** Use the household summaries n=70, mean=$9,312, s=$4,007 to construct a 95% interval. Explain the critical distribution and df.
4. **Proportion interval:** Use 396/900 for tee-time satisfaction. Compute a 95% Wald interval and state its conditions. Explain why it describes a population proportion rather than a range of binary observations.
5. **Planning:** Find the completed sample needed for E=.025 and p*=.50 at 95% confidence. Identify what the result does and does not guarantee.
6. **Communication:** Correct these claims: “95% of salaries lie in the mean confidence interval”; “n>30 means use z”; “a huge volunteer sample removes bias”; “the CLT makes the population normal.”
7. **Evidence limits:** List which inputs are published summaries, which computations rely on assumptions, and what raw data or design evidence would be required to assess those assumptions.

Bridge to hypothesis testing

An interval shows plausible parameter values under a procedure and its assumptions. The next module evaluates a specified null claim. Retain the same target parameter, sampling conditions, and distinction between practical importance and statistical uncertainty.

Return to the module overview.